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SYSTEM AND METHOD FOR BAG CONTROL IN A RECEPTACLE

Abstract

A receptacle is configured to receive a bag, such as a trash bag. The receptacle includes an outer shell, a plunger, and a fan. The plunger is moveable relative to the outer shell. The fan is configured to be rotated when the plunger is moved relative to the outer shell. Rotation of the fan expels air from the receptacle and sucks the bag open within the receptacle.

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Background/Summary

FIELD

[0001] This application relates to the field of garbage disposal and particularly to trash receptacles.

BACKGROUND

[0002] One of the more popular trash cans is the kitchen stainless steel can, with a black plastic “inner bucket”. Users remove the bucket from the outer can, then remove the full bag of trash. The inner bucket makes cleaning easier, contains the bag while it is carried outside, and is designed to help trash bags tuck in, for better fit around the upper lip.

[0003] However, when the trash is heavy or compacted, it can push against the bucket walls, which can cause both friction and suction, as the bag is being pulled out. Users may need one hand to hold down the bucket, while using the other hand to pull the bag up and out. This can be inconvenient, awkward and strenuous. It can also cause holes, rips or complete failure of the bag.

[0004] When placing a new bag in the inner bucket, the bag often does not fully open, and a virtual airlock tends to keep the bag inwardly billowed, even as trash is placed inside. Lightweight trash often stays on top, leaving little or no room for more.

[0005] Over many years, various mechanisms have been invented to address some of these issues, but none have had notable success, for a variety of reasons.

SUMMARY

[0006] A receptacle with fan is disclosed herein. The receptacle configured to receive a bag, such as a trash bag. The receptacle includes an outer shell, a plunger, and the fan. The plunger is moveable relative to the outer shell. The fan is configured to be rotated when the plunger is moved relative to the outer shell. Rotation of the fan expels air from the receptacle and sucks the bag open within the receptacle.

[0007] The above-described features and advantages, as well as others, will become more readily apparent to those of ordinary skill in the art by reference to the following detailed description and accompanying drawings. While it would be desirable to provide a system and method for bag control in a receptacle that provides one or more of these or other advantageous features as may be apparent to those reviewing this disclosure, the teachings disclosed herein extend to those embodiments which fall within the scope of the appended claims, regardless of whether they include or accomplish one or more of the advantages or features mentioned herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a rear view of a receptacle including an outer shell with fan and a bag support in the form of an inner bucket, the bag support configured for placement within the outer shell;

[0009] FIG. 2A is a lower perspective view of a rack-and-pinion arrangement with rack catch for the receptacle of FIG. 1;

[0010] FIG. 2B is a side view of the rack-and-pinion arrangement of FIG. 2A;

[0011] FIG. 3A is a plan view of a gear arrangement within a gearbox of the receptacle of FIG. 1;

[0012] FIG. 3B is a side perspective view of the gear arrangement of FIG. 3A, the gear arrangement connected to a fan;

[0013] FIG. 3C is a side view of the gearbox of FIG. 3A in isolation from the gears; FIG. 4 is a side view of the fan of FIG. 1;

[0014] FIG. 5A is a front view of an alternative embodiment of the inner bucket of FIG. 1;

[0015] FIG. 5B is a front view of another alternative embodiment of the inner bucket of FIG. 1;

[0016] FIG. 6 is a side perspective view of an alternative embodiment of the gearbox of FIG. 3A wherein a drive belt connects the gearbox to the fan;

[0017] FIG. 7A is a side perspective view of an alternative embodiment of the bag support of FIG. 1 wherein the bag support is provided by an inner platform;

[0018] FIG. 7B is a bottom perspective view of the platform of FIG. 7A;

[0019] FIG. 7C is a rear perspective view of the platform of FIG. 7A connected to a plunger, gearbox and fan;

[0020] FIG. 8 shows the plunger of FIG. 7C arranged on an exterior of a receptacle; and

[0021] FIG. 9 shows an alternative embodiment of the platform of FIG. 7A.

DESCRIPTION

[0022] The present disclosure solves these issues of friction, suction, airlock and bag under-inflation in novel ways, as described herein. FIG. 1 illustrates a fan **10**, mounted in the back wall of a receptacle having an outer shell **11** (which is also referred to herein as an “outer can”), and a bag support (e.g., an inner bucket **12**) configured to receive or otherwise engage a bag and be positioned within the outer can **11**. The fan **10** draws air outwardly or otherwise expels air from the outer can **11**. When the inner bucket **12**, with an already inserted trash bag **14** is put into place within the outer can **11**, the upper rim **15** of the inner bucket **12** forms an approximate seal with the upper rim **16** of the outer can **11**. So when the fan **10** is running, a vacuum forms in the space between the inner bucket **12** and the outer can **11**. But since there are holes **13** in or near the bottom of the inner bucket **12**, the trash bag **14** is pulled down and opened by the vacuum. In at least one embodiment, the fan **10** is located in the back of the outer can **11**, but it may be positioned elsewhere, for example at the bottom of the outer can **11**, under the inner bucket **12**.

[0023] The fan can be powered by batteries or electricity from an outlet, and be switched on and off manually, or by limit switches, photo-sensors, timers or other means. But these require things many consumers do not want, like batteries, motors and power cords. Embodiments of the present disclosure aim to be simpler, more efficient, more ecological, less costly, and more durable, by using human power, via a mechanism requiring very little effort, and almost no extra time.

[0024] FIGS. 2A and 2B illustrate two views of the fan drive mechanism **20** of the present disclosure. FIG. 2A is a front view, and FIG. 2B is a side view. When users have placed a new trash bag **14** into the inner bucket **12**, and then insert the inner bucket **12** into the outer can **11**, the inner bucket **12** first lands on the rack catch **21**, and begins pushing it down. This lowers the rack catch **21**, and also its rigidly attached rack **22**, which mechanically engages a gear arrangement within a gearbox **23**. As rack **22** slides lower within rack channel **24**, it continues to drive the first gear **31** (i.e., the pinion gear of a rack-and-pinion arrangement), and thus also gears **32**, **33**, and **34**, as shown in FIGS. 3A and 3B. Rack **22** engages the first gear **31** at point **35**. The final gear in gearbox **23** is fourth gear **34**, which is attached to the fan **10** by the drive shaft **38**. As the rack **22** lowers, it drives power through the gearbox **23** to propel the fan **10** to expel air from the outer can **11**. The weight of the inner bucket **12** may not be adequate to drive the fan **10** at a sufficient speed to expel enough air, so the user may push down on the inner bucket **12**, to achieve the desired fan speed.

[0025] The user need only push the inner bucket **12** down a few inches, to drive the fan **10** at high speed, due to the gear ratios in the gearbox **23**. Since the user is already manually inserting the inner bucket **12**, this short push requires almost no extra time, and little effort. In most prototype embodiments, the user presses down the inner bucket for approximately 1 second. During that time, the fan **10** is quickly expelling air. However, there is air leakage into the outer can **11**, since the upper rim **15** of the inner bucket **12** is not yet seated against the upper rim **16** of the outer can **11**. Therefore, the fan **10** is unable to fully inflate the bag.

[0026] But once the inner bucket **12** has been fully inserted, very little air can enter the outer can **11** from around the upper rim **15** of the inner bucket **12**. As the fan **10** continues to run on momentum, it expels air rapidly, pulling the vacuum sufficient to suck open the trash bag **14** (i.e., inflate the bag) inside the inner bucket **12**. Air holes **13**, in or near the bottom of the inner bucket **12**, allow the air surrounding the trash bag **14** to be pulled out, through the outer can **11**, and expelled, thereby pulling the bag **14** down and open against the walls of the inner bucket **12**. Next, the fan continues to run until its momentum dies from friction and air resistance.

[0027] For the fan **10** to continue running on momentum, it disengages from the rack **22**, which

stops moving upon fullinsertion of the inner bucket **12**. This disengagement occurs within the gearbox **23**, as shown in FIGS. **3A** and **3C**. FIG. **3C** illustrates the gearbox **23** without the gears. Three holes **36** retain the shafts of first gear **31**, third gear **33**, and fourth gear **34** in a fixed location. The shaft of the second gear **32** slides within slot **37**, allowing second gear **32** to engage or disengage the third gear **33**. When the rack **22** is pushed down, it drives the first gear **31** to rotate clockwise, which drives the sliding gear **32** up, along the slot **37** to engage the third gear **33**. The third gear **33** drives the fourth gear **34**, which drives the fan **10** via the drive shaft **38**. As the fan **10** continues on momentum, so does the fourth gear **34** and the third gear **33**. This continued rotation of third gear **33**, pushes the second gear **32** to move back down the slot **37** until the third gear **33** disengages the second gear **32**. The mostly vertical arrangement of gears aids this disengagement, as gravity helps prevent unwanted movement of second gear **32** back up toward third gear **33**.

[0028] To ensure the fan **10** momentum is sufficient to expel enough air, the fan **10** may be weighted and the gearbox **23** is designed for a reasonable downward force applied to the inner bucket **12**, by the user as described above. In one embodiment, the fan **10**, shown again in FIG. **4**, has weights **41** attached near the outer edge of each fan blade **42**, to increase angular momentum. This may also be accomplished with a flywheel attached to the hub of the fan blade assembly, or a larger diameter weighted ring attached across the outer edges of each fan blade, or by other means. [0029] When the user removes the inner bucket **12**, the rack catch **21** is no longer weighed down by the inner bucket **12**, and the rack extension spring **25** can then pull the rack **22** up and along the rack channel **24**, to its reset position. Since the second gear **32** is still disengaged from the third gear **33**, raising the rack **22** does not drive the fan. Instead, it drives the first gear **31** counterclockwise, which, along with gravity, keeps the second gear **32** pushed away from the 3rd gear **33**. In at least one embodiment, the gears and rack are not heavy, and are designed to move with little friction, so the rack extension spring **25** can be light duty, thus adding very little resistance to a user inserting the inner bucket **12**.

[0030] FIGS. **5A** and **5B** illustrate additional inner bucket **12** features of the present disclosure. At the bottom of the inner bucket **12**, on each side is a concave foothold **50**, of FIG. **5A**, where users may step to hold down the inner bucket **12**, while pulling the full trash bag **14** up and out of the inner bucket **12**. This stabilizes the inner bucket **12**, and firmly holds it to the floor, while allowing the user to use both hands to remove the trash bag **14**. Another embodiment has only one foothold **51**, of FIG. **5B**, positioned front and center at the base of the inner bucket **12**, and also is taller for easier foot insertion. Another embodiment may have both of these options (foothold **51** and footholds **50**) for users to choose from. Other embodiments have different foothold locations, and different foothold sizes and shapes.

[0031] The air holes **13**, in or near the bottom of the inner bucket **12**, allow air to freely enter, immediately relieving any suction that may start to develop as the trash bag **14** is removed. To help contain any trash leakage, at least one embodiment positions the air holes **13** not in the bottom of the inner bucket **12**, but in the side, near the bottom. This location aids in both insertion and extraction of trash bag **14**.

[0032] Many trash cans have lid hinges that protrude from the back of the can, which prevents the lower part of the can from sitting flush against a wall or cabinet. This gap can provide room for the fan **10** and the air it will expel. One embodiment of the current disclosure has the fan **10** positioned entirely within the body of the outer can **11**, which may be associated with the inner bucket **12** being contoured around the fan **10** and gearbox **23**. Another embodiment has the fan **10** mounted on the outside of the outer can **11**. Still other embodiments have the fan **10** positioned in between—for example, mostly inside or mostly outside the outer can **11**. For safety, the fan **10** is covered by an inflexible screen or grate. Bladeless fans may also be used to increase safety.

[0033] Another embodiment uses a space underneath the inner bucket **12** to mount the fan **10**, which may be driven by a belt, worm gear, drive shaft or other means extending from the gearbox **23** to the fan **10**. In this embodiment, the lower portion of the outer can **11** has vents, through which

the fan **10** expels the air from the outer can **11**.

[0034] The high fluidity of air allows for many fan placement options. For better dispersion of expelled air, some placement options may benefit from the use of centrifugal fans or diagonal fans instead of the axial fan shown in FIG. **4**.

[0035] In transferring power to the fan **10**, another embodiment uses pulleys and belts, such as V-belts or timing belts, or chains or cables instead of a rack and gears. Another embodiment uses one or more freely spinning cylinders with spiraled slots, wherein a key inserted into the slot is driven downward during insertion of the inner-bucket, thereby spinning the slotted cylinder, to provide power for the fan **10**. The pitch of the spiraled slots may be varied, to impart more control to fan acceleration.

[0036] In the design shown in FIGS. **2B** and **3B**, the fan **10** is directly attached to the final gear **34** in the gearbox **23**. This configuration adds thickness to the assembly which may cause the fan **10** to stick out of the trash can undesirably far. FIG. **6** illustrates one embodiment designed to reduce the total thickness, by stacking the fan **61** and gearbox, shown here as enclosed gearbox **62**, using a timing belt **63** to transfer power from the enclosed gearbox **62** to the fan **61**. Instead of a timing belt, a V-belt, chain, driveshaft, worm gear or other means may be used. The design in FIG. **6** also removes the enclosed gearbox **62** from the path of airflow, which can increase efficiency. A fully enclosed gearbox is quieter, keeps the gears cleaner, and helps contain gear lubricant. But if the open gearbox **23** of FIGS. **3A-3C** were fully enclosed it might hinder airflow. This may be another advantage of the configuration of FIG. **6**.

[0037] In other embodiments, the rack catch **21** shown in FIGS. **2A** and **2B**, can be replaced by a platform, or one or more cross-members, such as bars, cables, wires, or beams, which span fully or partially from one side to another, and may be fully or partially supported by, attached to, guided by or mounted on supports such as springs, cables, racks, screws rails or gears.

[0038] FIGS. **7A-7C** show three views of another embodiment, which is designed for trash cans having no inner bucket. A platform **70** is made to fit into the bottom of the can, with the fit being tight enough to create a virtual air seal with the can. A gasket may be used to enhance the seal. Ribs **71** on the underside of the platform **70** hold the platform **70** up off the bottom of the trash can (e.g., between one inch and three inches above the bottom of the trash can), thus creating a cavity between the platform **70** and the bottom of the trash can. The ribs **71** are substantial to support the weight of heavy trash or items that may be placed on the platform **70**. The ribs **71** are formed and arranged to leave channels **72** through the cavity under the platform **70**, which allow air to flow freely throughout the cavity. Holes **73** in the platform **70** allow air to flow from the trash can into the cavity under the platform **70**. Arising from one side of the platform **70**, is an air duct **74**, leading to the exhaust fan **75**.

[0039] There may be numerous acceptable designs for the ribs **71** and channels **72**. For example, some designs might channel all air from the holes toward a large central channel which leads to the air duct **74**. But air is extremely fluid, so as long as it can freely flow from any hole to the air duct **74**, the precise airflow pathways may be of little consequence. Of more consequence is the hole pattern, designed to pull the bag down and toward the outer walls of the can. Ribs positioned vertically up the walls of the can seem to aid in fully opening the bag, but only marginally. Various hole patterns work well, even without the vertical ribs, and there may be many hole patterns that work well. The pattern shown in FIG. **7A** has most holes lining the outer rim, to pull the bag down and outward.

[0040] The embodiment shown in FIGS. **7A-8** use the same mechanical system (rack, gearbox, and fan) as that of FIGS. **2A-3B** and FIG. **6**. But when there is no inner bucket for users to push down, the system is powered in some other way. In this embodiment, a hand plunger **76** is attached to the rack (inside housing **77**), or may be manufactured as rack and plunger being one piece. Just as the rack of FIGS. **2A** and **2B** is mechanically attached to gearbox **23** of **3A** and **3B**, the plunger **76** drives power through the gearbox **78** and to the fan **75**. When the plunger **76** is completely down in

its resting position, it appears to the user as a small knob **80** on the end of the plunger, as shown in FIG. **8**. The user pulls up the knob **80**, thereby pulling up the plunger **76**. This is easy, since the 2nd gear **32** is disengaged from 3rd gear **33** (referring to the gearbox **23** of FIGS. **2A-3B**). The user then pushes the plunger **76** down, to drive the fan, which pulls air from the space between the bag and the can, the air then traveling down through holes **73** into the cavity below the platform **70**, and through the channels **72** between the ribs **71**, then through the duct **74** to the fan **75**, which expels the air. This embodiment could also be used with cans having inner buckets, but perhaps with less added benefit.

[0041] The assembly that includes the air duct **74**, gearbox **78**, rack housing **77** and plunger **76** may be built into an outer wall of the trash can, or be contained completely within it, or be mounted onto the outside of it, or something in between.

[0042] There are several ways to minimize interference of the knob **80** and plunger **76** with the upper rim of the trash bag. For example, the knob **80** and plunger **76** can be positioned just outside the lip of the trash can. Or the lip of the trash can, can be made concave at that point, to circumvent the knob **80** and plunger **76**. In another embodiment, the plunger **76** slides in a slot or C-channel mounted on the outside of the trash can, with the C-channel extending only to just below the lip of the trash can. This allows the knob **80** to protrude horizontally from the rack, further distancing it from the trash bag. Also, the plunger can be mounted closer to the bottom and can be operated either by hand or by foot. With some trash can designs, this may work better with a spring, which always returns the rack and plunger **76** to the up position, ready to plunge downward, without having to manually raise the plunger **76** each time. This may work better in conjunction with a dashpot for a “soft reset” of the plunger **76**. This embodiment can be applied to trash cans with inner buckets too, wherein the inner bucket does not engage the mechanism. Instead, the user inserts the bucket completely, and then pulls down, or pushes down a plunger mounted within or onto the side of the trash can. This could be positioned for hand use or for foot use. The plunger can be located at the top, and be pulled up manually each time, or be mounted lower, and be automatically pulled up by a spring, along with a dashpot for a “soft reset”. Alternately, instead of an extension spring to pull up a rack or plunger from above, a compression spring can be used to push it up from below. This allows more placement options for the rack or plunger **76**.

[0043] The platform **70** can be designed to permanently attach to the trash can or attached with easily removed fasteners to aid in cleaning. Or it can be designed to simply drop into place, or snap into place, for even easier cleaning. If there is any trash fluid leakage, the platform design shown in FIGS. **7A-7C** can allow some of the leakage into the cavity under the platform. To help minimize this issue, the platform can be designed with the air holes **73** positioned above the central base of the platform, as shown in FIG. **9**. Each air hole **73** may also have a lip around its perimeter to help channel any leakage away from the hole.

[0044] In some bucket-less cans the plunger can be made shorter than the rack of an equivalent can that does have an inner bucket. This is because while the bucket is being pushed down to power up the fan, mass air is leaking around the bucket into the can. But with bucket-less cans, there is no such initial period of mass air leakage. Instead, the bag begins to pull open when the user starts pushing down the plunger. If some trash can designs need additional rack length to attain the power needed, a flexible rack can be used, wherein the flexible rack follows inside a curved sleeve or c-channel to curl under or along the base of the can.

[0045] Some products have features that users can opt to disengage or turn off. With the embodiments described herein, this could be done in any of several ways. For example, the top of the rack ex-tension spring can be attached to a releasable fixture, such that users can switch it to unlock, whereupon the releasable fixture will slide down a channel to the spring's fully relaxed position. To reset it, the users slides the fixture back to its original position, and switches it to lock. A locking lever can be used instead, which would perform both tasks-resetting and locking the fixture. With the plunger embodiment, a switch brake can be mounted on the plunger to lock it in

place.

[0046] Many of the embodiments herein disclosed can be retrofitted to existing trash cans, including those with inner buckets, and those without. However, integrating these embodiments into new trash can designs may result in better products and better user experiences.

[0047] The present disclosure can be used with many different sizes and designs of cans, bins and containers of all types, including but not limited to: trash cans, recycle cans, storage containers, bio-hazard cans, supply bins, such as industrial bins containing parts and materials for assembly.

[0048] The foregoing detailed description of one or more embodiments of the trash can with fan has been presented herein by way of example only and not limitation. It will be recognized that there are advantages to certain individual features and functions described herein that may be obtained without incorporating other features and functions described herein. Moreover, it will be recognized that various alternatives, modifications, variations, or improvements of the above-disclosed embodiments and other features and functions, or alternatives thereof, may be desirably combined into many other different embodiments, systems or applications. Presently unforeseen or unanticipated alternatives, modifications, variations, or improvements therein may be subsequently made by those skilled in the art which are also intended to be encompassed by any appended claims. Therefore, the spirit and scope of any eventually appended claims should not be limited to the description of the embodiments contained herein.

Claims

1. A receptacle configured to receive a bag, the receptacle comprising: an outer shell; a plunger moveable relative to the outer shell; and a fan configured to be operated when the plunger is moved relative to the outer shell.
2. The receptacle of claim 1, wherein rotation of the fan expels air outwardly from the receptacle.
3. The receptacle of claim 2, further comprising a bag support moveably positioned within the outer shell.
4. The receptacle of claim 3, further comprising a gear arrangement, the plunger configured to engage the bag support and the gear arrangement.
5. The receptacle of claim 4, wherein movement of the plunger in a downward direction causes the gear arrangement to rotate the fan and expel air outwardly from the receptacle.
6. The receptacle of claim 5, wherein the plunger is a rack of a rack-and-pinion arrangement and the gear arrangement includes a pinion of the rack-and-pinion arrangement.
7. The receptacle of claim 6, wherein the bag support is an inner bucket including at a plurality of holes at a bottom portion of the bucket.
8. The receptacle of claim 7, wherein the rack is connected to a rack catch positioned within the outer shell and configured to engage the inner bucket.
9. The receptacle of claim 1, further comprising a bag support provided by a platform above a bottom of the receptacle with a plurality of holes formed in the platform, wherein the fan is configured to direct air through the holes in the platform when the plunger is moved relative to the outer shell.
10. The receptacle of claim 9 further comprising ribs on an underside of the platform, wherein the ribs distance the platform above the bottom of the receptacle such that a cavity is formed between the platform and the bottom of the receptacle, the fan configured to direct air through the holes in the platform and into the cavity when the plunger is moved relative to the outer shell.
11. A system for installing a bag in a receptacle, the receptacle comprising: an outer shell configured to receive the bag therein; a mechanical actuator that is moveable relative to the outer shell; and a fan configured to be rotated when the actuator is moved relative to the outer shell.
12. The system of claim 11 wherein the mechanical actuator is a plunger.
13. The system of claim 12 further comprising a bag support removably positioned within the outer

shell, the bag support configured to receive the bag.

14. The system of claim 13 wherein the bag support moves the plunger when the bag support is inserted into the outer shell.

15. The system of claim 12 wherein the plunger includes an end configured to be grasped by a human hand on an exterior of the outer shell.

16. The system of claim 12 further comprising a gear arrangement coupling the plunger to the fan, the gear arrangement including a pinion and the plunger serving as a rack of a rack-and-pinion system.

17. The system of claim 12 further comprising a platform positioned within the outer shell, the platform positioned above a bottom of the outer shell such that a cavity is formed between the platform and the bottom of the outer shell.

18. The system of claim 17 further comprising ribs connected to the platform and distancing the platform from the bottom of the outer shell.

19. The system of claim 18 wherein the platform includes a plurality of holes such that rotation of the fan is configured to direct air through the plurality of holes.

20. A method of controlling a bag in a receptacle comprising: positioning the bag in the receptacle; and moving an actuator in order to operate a fan and draw air through the receptacle such that the bag is inflated within the receptacle.
